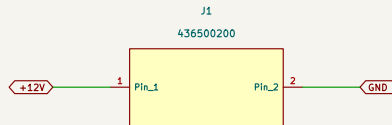
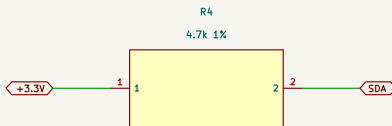
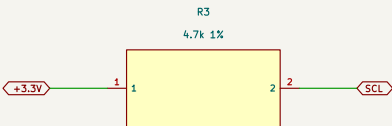
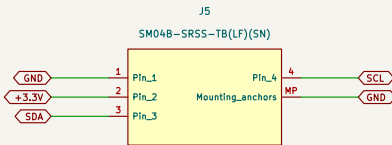
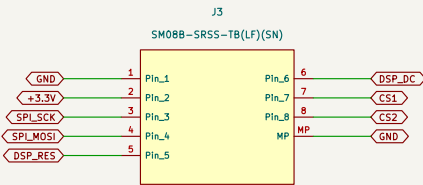
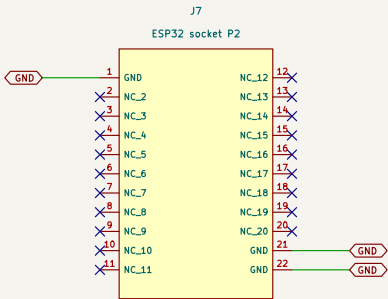
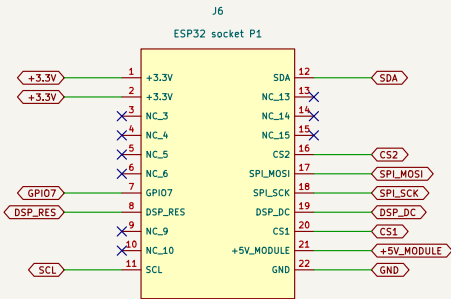
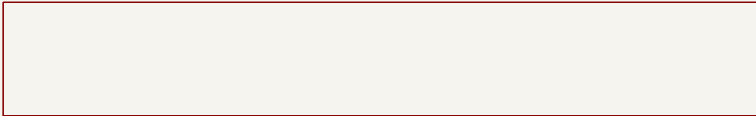


Controller and connectors



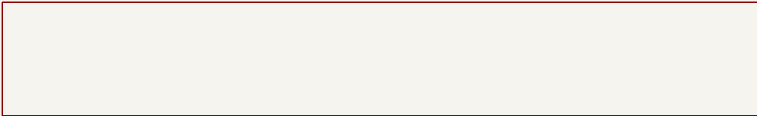
Waveshare ESP32-S3-DEV-KIT-N16R8-M (SKU 28836), separately purchased.
Sockets: P1=J6, P2=J7. 2.54mm pitch, 22.86mm row spacing.
All carrier parts top side. Remove J8 shunt before USB connection.

Accessory protection



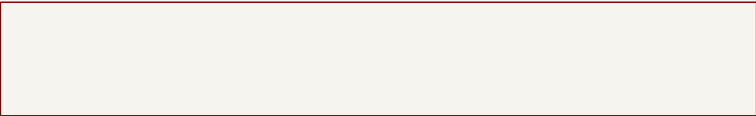
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5V regulator and USB isolation



File: sheet3.kicad_sch

Clock and local sensors



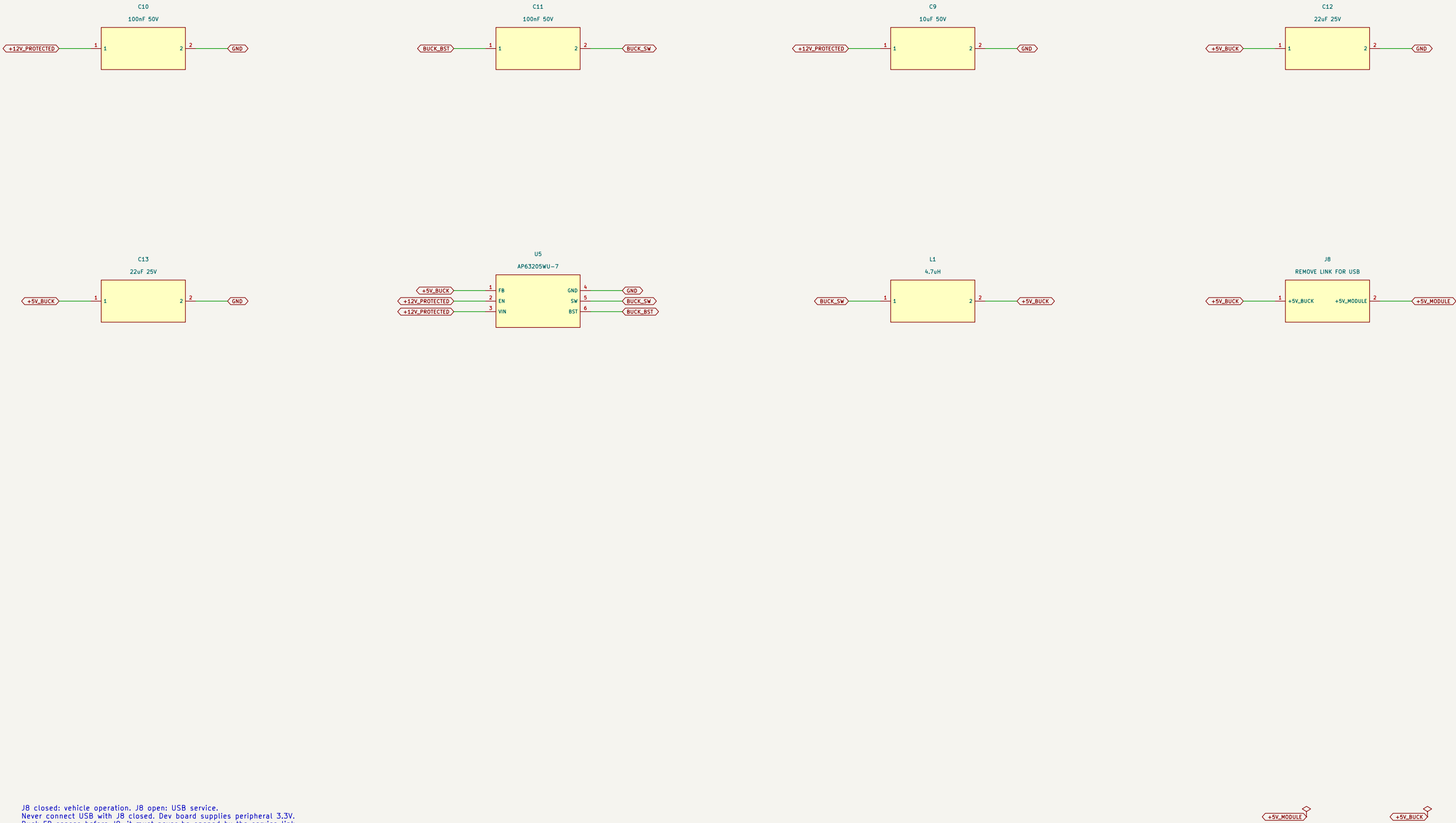
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Accessory protection

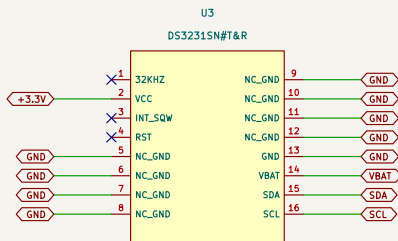
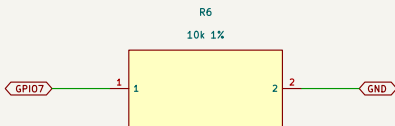
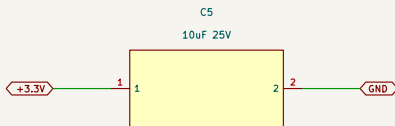
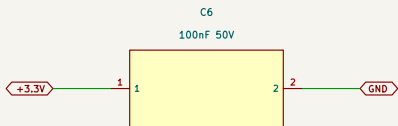
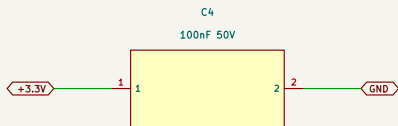
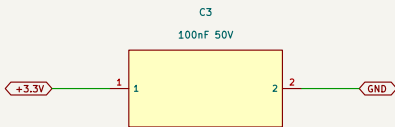
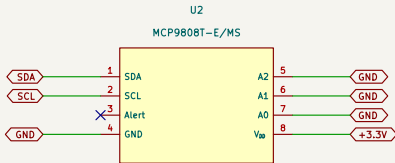
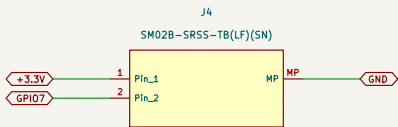
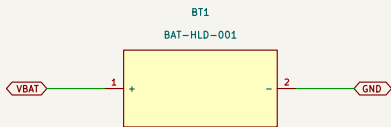


RTN and exposed pad are isolated from GND.
TVS pair has common anodes. Fused switched accessory branch required.
Prototype protection envelope and physical tests remain as documented.

5V regulator and USB isolation



Clock and local sensors



DS3231: NC pins 5-12 tied to GND per manufacturer.
CR2032 is non-rechargeable; no charging circuit. RTC RST is unused.
BME280 readings remain separate from the local board temperature.

